Microsoft Azure Fundamentals (AZ-900)

Advanced Practice Questions & Answer Key

1. An enterprise migrates a legacy monolithic application using a 'lift-and-shift' approach to Azure Virtual Machines. Under the shared responsibility model, which of the following layers remains strictly the customer's accountability?

Correct Answer: Guest operating system patches, application middleware, and identity access control.

2. Which cloud deployment model presents the highest level of regulatory compliance complexity by intentionally integrating multi-tenant public utility scalability with a dedicated, isolated on-premises or hosted private infrastructure?

Correct Answer: Hybrid cloud

3. A financial institution requires an architecture where their database storage tier is completely isolated from public access, yet needs to securely communicate with a web app deployed in a different subnet. Which Azure networking feature best provides this secure internal connectivity?

Correct Answer: Virtual Network Service Endpoints

4. An organization wants to restrict resource creation so that developers can only deploy resources within specific geographic regions (e.g., North Europe) to comply with data residency laws. Which Azure governance tool should be enforced?

Correct Answer: Azure Policy

5. When designing a mission-critical workload for high availability across an Azure region, how are Availability Zones structurally isolated?

Correct Answer: They consist of separate physical datacenters with independent power, cooling, and networking.

6. What is the primary architectural purpose of utilizing Azure Resource Manager (ARM) templates or Bicep files?

Correct Answer: To enable Infrastructure as Code (IaC) for consistent, repeatable, and declarative deployments.

7. An administrator wants to apply a 'CanNotDelete' resource lock to a production database to prevent accidental purging. Who has the authority to delete the resource while this lock is active?

Correct Answer: No user, including Owners and Global Administrators, can delete the resource until the lock is removed.

8. Which Azure pricing model offers significant discounts (up to 72% compared to pay-as-you-go) in exchange for committing to a 1-year or 3-year term on specific compute services?

Correct Answer: Azure Reserved Instances

9. What is the primary function of Azure Management Groups?

Correct Answer: To provide a hierarchical scope above subscriptions for applying policies, access control, and compliance at scale.

10. Which Azure support plan provides 24/7 access to technical support engineers with a response time of less than 1 hour for critical business-impact (Severity A) incidents?

Correct Answer: Professional Direct Support

11. An administrator needs to securely connect an on-premises datacenter directly to an Azure Virtual Network bypassing the public internet entirely, ensuring high throughput and reliable latency. Which service should be configured?

Correct Answer: Azure ExpressRoute

12. What is the primary role of the Azure Global Infrastructure component known as a 'Geography'?

Correct Answer: A discrete market, typically spanning two or more countries, that preserves data residency and compliance boundaries.

13. Which cloud service model shifts the responsibility of managing runtime frameworks, operating systems, and middleware entirely to the cloud provider, allowing developers to focus solely on writing application code?

Correct Answer: Platform as a Service (PaaS)

14. An organization wants to track cloud costs down to specific departments, projects, and environments by attaching descriptive metadata labels to resources. What Azure feature enables this?

Correct Answer: Azure Tags

15. Which Azure identity feature enables organizations to grant users temporary, just-in-time (JIT) privileged access to Azure resources to mitigate the risk of excessive standing permissions?

Correct Answer: Microsoft Entra Privileged Identity Management (PIM)

16. What is the primary function of Azure Defender for Cloud (formerly Azure Security Center)?

Correct Answer: To provide unified security management and advanced threat protection across hybrid cloud workloads.

17. An application needs to scale out automatically based on incoming CPU utilization metrics during peak traffic hours and scale back in when demand subsides. Which Azure feature provides this capability natively for PaaS and compute tiers?

Correct Answer: Azure Autoscale

18. Which Azure storage tier is optimized for storing data that is rarely accessed and must be retained for at least 180 days with lower storage costs but higher retrieval latency?

Correct Answer: Archive tier

19. What is the primary purpose of Azure Bastion in a secure virtual network architecture?

Correct Answer: To provide secure, seamless RDP and SSH connectivity directly to virtual machines over SSL without exposing public IP addresses.

20. When creating an Azure resource, what is the mandatory foundational container scope required to organize and bill that resource?

Correct Answer: Resource Group

21. Which Azure tool provides personalized recommendations regarding cost optimization, security posture, reliability, operational excellence, and performance?

Correct Answer: Azure Advisor

22. What is the main characteristic of serverless computing services in Azure, such as Azure Functions?

Correct Answer: Infrastructure management is abstracted away, scaling is automatic and elastic, and users pay only for the exact compute resources consumed during execution.

23. Which cloud deployment model refers to cloud resources owned and operated exclusively by a single organization, hosted either on-premises or by a third-party datacenter provider?

Correct Answer: Private cloud

24. How does Azure ensure high availability across an Azure region when deploying virtual machines using Fault Domains?

Correct Answer: Fault domains define hardware racks that share common power and network switch resources, ensuring VMs are separated across different racks.

25. Which service should an enterprise use to centrally collect, query, and analyze log telemetry and metrics generated across all Azure resources and hybrid VMs?

Correct Answer: Azure Monitor (Log Analytics Workspace)

26. What is the primary function of Microsoft Entra External ID (formerly Azure AD B2C)?

Correct Answer: To provide identity management and secure authentication for consumer-facing and external application users.

27. An administrator wants to ensure that all newly created storage accounts automatically have public blob access disabled by default across an entire subscription. What is the most efficient Azure governance feature to enforce this?

Correct Answer: Azure Policy with a 'Deny' or 'Audit' effect

28. Which Azure feature enables global load balancing of incoming HTTP/HTTPS web traffic across multiple regions while providing web application firewall (WAF) protection and SSL offloading at the edge?

Correct Answer: Azure Front Door

29. What is the primary benefit of utilizing Azure PaaS database services, such as Azure SQL Database, compared to running a database on an IaaS Virtual Machine?

Correct Answer: Automated patching, built-in high availability, automated backups, and elastic scaling managed by Microsoft.

30. An enterprise wants to analyze historical cloud spending trends, forecast future budgets, and set up automated alerts when expenditures exceed specific thresholds. Which tool should they use?

Correct Answer: Azure Cost Management + Billing